

Ziseok Lee



Personal Website: <https://ziseoklee.github.io/>
Google Scholar: https://scholar.google.com/citations?user=C3x_tewAAAAJ&hl=en
Contact: ziseoklee@snu.ac.kr



Integrated Masters with PhD Program
Department of Biomedical Sciences
Seoul National University

Research Interests

Generative Models, AI for Science, Computer Vision

Education

Seoul National University	2025.03 - present
Masters-PhD Program in Biomedical Sciences advised by Prof. Kyungsu Kim	
Seoul National University	2021.03 - 2025.02
B.S. in Mathematical Sciences (Major), B.S. in Computer Science and Engineering (Double Major)	
GPA 4.22/4.30 (Summa Cum Laude)	

Research Experience

▼ Student Researcher at AIBL	2024.10 - present
Probabilistic Generative Models for Biomolecule Generation and Drug Design	
Designed stochastic interpolant-based sampling methods for drug optimization and integrating physics-aware constraints for structure-based drug design.	
Derived and implemented the ACE framework for multi-expert diffusion steering, fixing Marginal Path Collapse via the Path Existence Criterion.	
Authored a grant proposal and secured funding for a 3-year international joint research project between SNU and Harvard Medical School (Prof. Kwang-Soo Kim).	
Co-first authored multiple papers on applications of diffusion models in computer vision and molecular modeling.	
▼ Undergraduate Research Assistant at SNU MLLAB	2024.07 - 2024.09
Blackbox Optimization	
▼ Undergraduate Research Assistant at CTA Lab	2024.01 - 2024.08
NP-hard Subgraph Matching Algorithms with Safety Conditions for Reduced Candidate Space	
▼ Undergraduate Research Assistant at HYKE Group	2022.12 - 2023.07
Collective Motion in the Vicsek Particle Model, Hyperbolic Conservation Laws with Nonlocal Relaxation	

Awards & Scholarships

AI Seoul Tech Research Support Program (Seoul Future Foundation)	2026
AI Seoul Tech Graduate Scholarship (Seoul Scholarship Foundation)	2025
B.S. Degree Honors: Summa Cum Laude (Seoul National University)	2025
Presidential Science Scholarship (Korea Student Aid Foundation)	2023-2024

Dean's List (Department of Mathematical Sciences, SNU)	2023-2024
Professor Heo Sik Scholarship (Department of Mathematical Sciences, SNU)	2022
Gwanak Foundation Scholarship (Gwanak Foundation)	2021

Teaching Experience

▼ Teaching Assistant for [Data Mining and Machine Learning]	2026.03 - 2026.06
Assisted Professor Kyungsu Kim in teaching a third-year undergraduate course, supporting students through discussion sessions.	
▼ Teaching Assistant for [Introduction to Data Science]	2025.03 - 2025.06
Assisted Professor Kyungsu Kim in teaching a second-year undergraduate course, supporting students through discussion sessions.	
▼ Teaching Assistant for [Calculus 1]	2024.03 - 2024.06
Taught a course on Calculus 1. Was awarded Excellent TA and gave a case presentation.	
▼ Student-Directed Seminar [Understanding the Brain as a Complex System]	2023.09 - 2023.12
Organized and taught a student-directed seminar integrating theoretical neuroscience, network science, deep learning, biology, and psychology to understand the brain as a complex, dynamical, entangled system.	
▼ Peer Tutoring [Calculus Tutoring for First Year Students]	2023.03 - 2023.06
Taught a tutoring class for "Differential and Integral Calculus 1" during the Spring semester of 2023.	

Grants & Funding

▼ Strategic Hub for International Research Collaboration (SNU Office of Research Affairs)	2026 - 2028
Project Title: An International Research Hub for Developing a 4D Molecular Generative AI Platform to Control Dynamic Protein Interaction Networks.	
Funding: ~\$380k (3 Years)	
Description: Authored the grant proposal and secured funding for a 3-year international joint research project between SNU and Harvard Medical School (Prof. Kwang-Soo Kim).	
My Role: I lead the design of stochastic interpolant-based diffusion models for drug optimization and integrating physics-aware constraints for structure-based drug design.	

Publications

* Equal contribution, † Corresponding Author

Conference Papers

- [Co-first Author] *ISAC: Training-Free Instance-to-Semantic Attention Control for Improving Multi-Instance Generation*, Sanghyun Jo*, Wooyeol Lee*, Ziseok Lee*, Kyungsu Kim†, ECCV 2026 (2026)
- [Co-first & Co-corresponding Author] *On the Collapse of Generative Paths: A Criterion and Correction for Diffusion Steering*, Ziseok Lee*†, Minyeong Hwang*, Wooyeol Lee, Sanghyun Jo, Jihyung Ko, Young Bin Park, Jae-Mun Choi, Eunho Yang†, Kyungsu Kim†, ICML 2026 (2026)
- [Co-first Author] *TRACE: Your Diffusion Model is Secretly an Instance Edge Detector*, Sanghyun Jo*, Ziseok Lee*, Wooyeol Lee, Jonghyun Choi, Jaesik Park†, Kyungsu Kim†, ICLR 2026 (**Oral, top 1.1%**) (2026)
- [Co-first Author] *Early Timestep Zero-Shot Candidate Selection for Instruction-Guided Image Editing*, Joowon Kim*, Ziseok Lee*, Donghyeon Cho, Sanghyun Jo, Yeonsung Jung, Kyungsu Kim†, Eunho Yang†, ICCV 2025 (2025)

Workshop Papers

[Coauthor] *HybridLinker: Topology-Guided Posterior Sampling for Enhanced Diversity and Validity in 3D Molecular Linker Generation*, Minyeong Hwang, Ziseok Lee, Kwang-Soo Kim, Kyungsu Kim†, Eunho Yang†, ICML 2025 Generative AI and Biology (GenBio) Workshop (2025)

Preprints

[Co-first Author] *Value-Informed Schedule Trimming for Accelerated Sequential Monte Carlo Guidance in Discrete Diffusion*, Jaehyeon Kim*, Ziseok Lee*, Siyoung Choi, Kyungsu Kim†, (2026)

Academic Services

- [2026.05] Reviewer for NeurIPS 2026.
 - [2025.04] Assisted Professor Kyungsu Kim in reviewing submissions in the field of computer vision for ICCV 2025.
 - [2025.03] Assisted Professor Kyungsu Kim in reviewing two submissions in the field of flow matching for ICML 2025.
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Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript, HTML/CSS

Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV

Tools & Platforms: High-performance GPU server management, Linux system administration

Languages: Korean (native), English (fluent)

References

Dr. Kyungsu Kim (Primary advisor)

Professor, School of Transdisciplinary Innovations

Seoul National University

Email: kyskim@snu.ac.kr